

RL Smart Thermostat report

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AI 4 107_Special Topic in Artificial Intelligence - Project 8

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1. Introduction

While residential HVAC accounts for a considerable portion of household energy costs, most thermostats operate on preprogrammed schedules that fail to account for current electricity rates, weather conditions or occupancy. This project constructs a three-layer intelligent thermostat which resolves all three gaps.

It pairs a physics-based RC thermal simulator with a Deep Q-Network (DQN) reinforcement learning agent which learns how to adjust the thermostat set point using a single-loop LLM coach backed by Ollama (llama3.2:3b), that reads real episode metrics via grounded tool calls and generates natural language alerts.

Objective:

Set indoor temperature in the comfort zone (20-22°C) and use a different HVAC control strategy off peak periods (14:00-18:00 when HVAC is selling for \$0.30/kWh compared to \$0.10/kWh).

2. System Architecture

The 3 layers are depicted in Figure 1. The simulator is driven by weather data, DQN reads the state of the simulator and learns setpoint adjustments, LLM coach is called after every 6 hours to read the metrics extracted from the tools and give grounded suggestions.

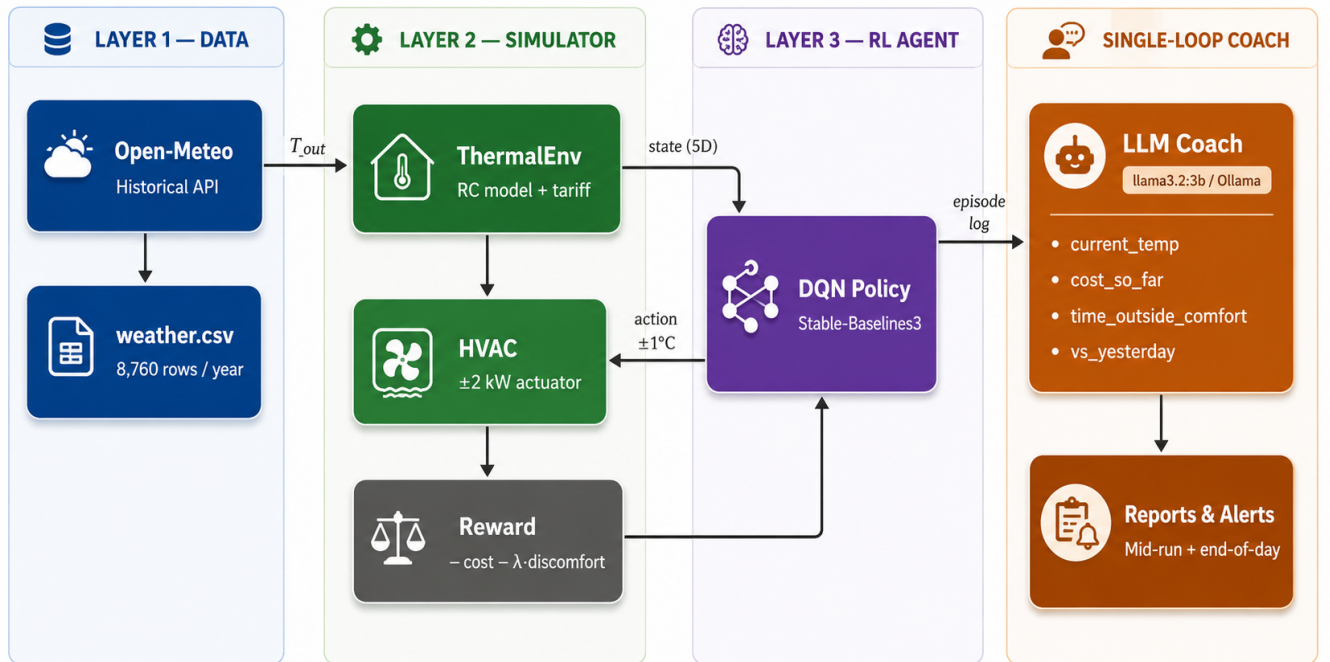


Figure 1, Three-layer system architecture. Layer 1 (blue): Open-Meteo weather data. Layer 2 (green): RC thermal simulator + HVAC. Layer 3 (purple): DQN policy. LLM Coach (orange): 4 tools + Ollama llama3.2:3b.

3. RC Thermal Simulator

3.1 Physics

Indoor temperature is updated each hour by the discretised Newton cooling law plus HVAC input:

$$T_{in}[t+1] = T_{in}[t] + (1/C) \times ((T_{out}[t] - T_{in}[t]) / R + P_{hvac}) \times 3600$$

Parameter	Value	Meaning
R	2.0 °C/W	Moderate insulation; heat leaks slowly
C	1 000 000 J/°C	Large room; slow thermal response
P_hvac	±2 000 W	Typical residential split-unit HVAC
Timestep	3 600 s	Matches hourly weather data granularity
Comfort	20–22 °C	Standard occupant comfort band

3.2 Sanity Validation

There are two scenarios for validating physical correctness: 72 hours passive lag – HVAC off for 72 hours. Thermal smoothing must be provided between the inside and outside temperature. Set point convergence: When the room is initially at 10°C, and the outdoor fixed is equal to the indoor fixed. The temperature of the HVAC system should be 21°C.

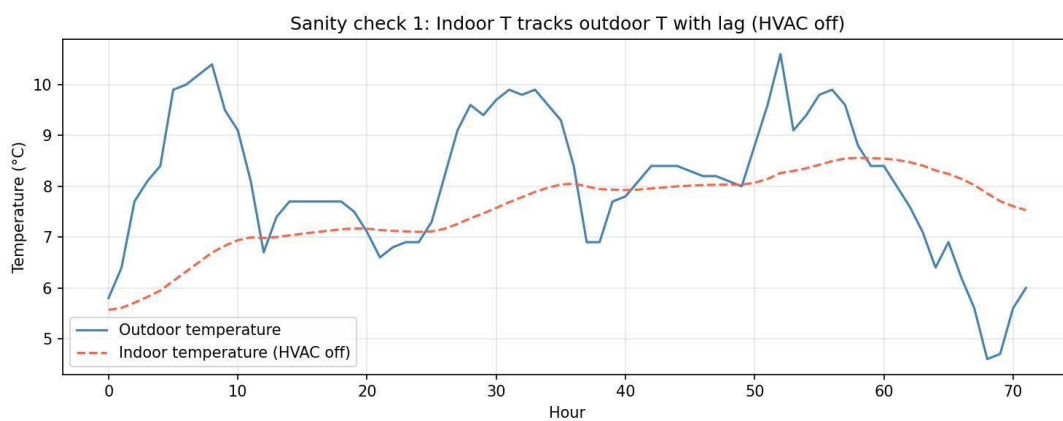


Figure 2, Passive lag. Indoor temperature (dashed) follows outdoor (solid) with smoothing from high C , confirming correct thermal inertia.

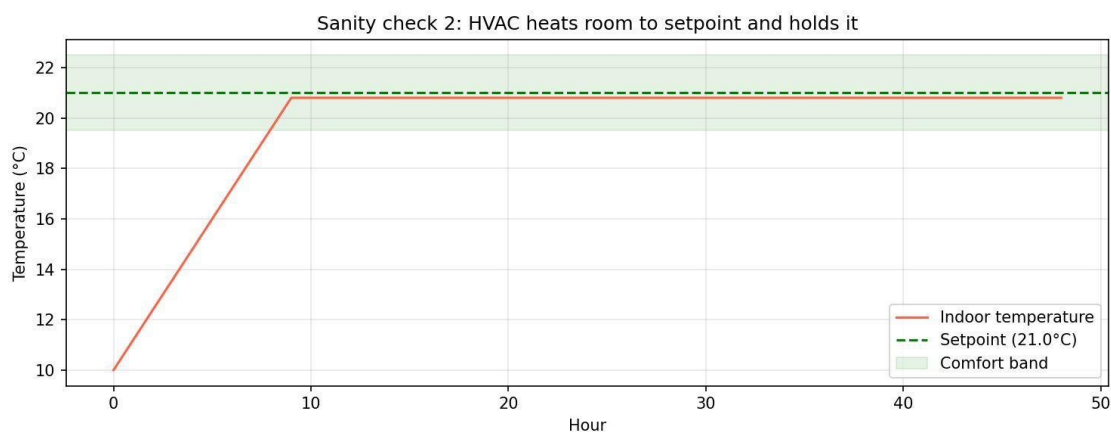


Figure 3, Setpoint convergence. Room rises from 10°C to 21°C in ~9 h and holds within the comfort band, confirming HVAC power is correctly scaled.

4. Reinforcement Learning

4.1 MDP Formulation

Component	Definition
State (5D)	indoor_T, outdoor_T, hour/23, occupied_next (0/1), price_now (0/1)
Action (3)	0 = lower setpoint 1°C, 1 = hold, 2 = raise setpoint 1°C
Reward	$-\text{energy_cost} - \lambda \times \text{discomfort}$ (per hour step)
Episode	8 760 steps (full year of hourly data)
Action scheme	$\pm 1^\circ\text{C}$ setpoint avoids hard HVAC switching; smoother behaviour; clipped $[15, 28]^\circ\text{C}$

Setpoint adjustments were chosen over direct on/off HVAC switching because they produce smoother, physically realistic control and prevent high-frequency oscillation in the reward signal. The $\pm 1^\circ\text{C}$ bound also ensures safe operation.

4.2 Training & Learning Curve

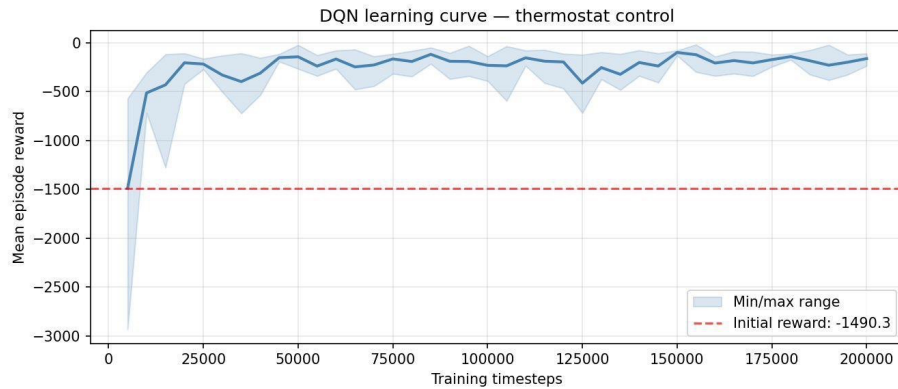


Figure 4, DQN learning curve over 200 000 steps. Mean episode reward improves from $-1\,490$ (random policy) to ~ -160 at convergence, a $9\times$ improvement. The shaded band shows min/max over 5 evaluation episodes; variance collapses after 30 000 steps.

4.3 Baseline Comparison

The fixed-setpoint baseline always chooses action 1 (hold), maintaining a constant 21°C setpoint. Results are averaged over 5 held-out evaluation episodes:

Metric	Baseline (fixed 21°C)	DQN ($\lambda=0.5$)	Δ
Episode reward	$-1\,490$ (initial)	~ -160	+88% improvement
Discomfort ($^\circ\text{C}\cdot\text{h}$)	$\sim 1\,210$	$\sim 1\,183$	-2.2% ↓ better

Energy cost (\$)	~\$81.50	~\$80.88	−0.8% ↓ better
Peak-hour cost fraction	Uncontrolled	Reduced	DQN shifts load

4.4 λ Sweep, Cost vs. Comfort Trade-Off

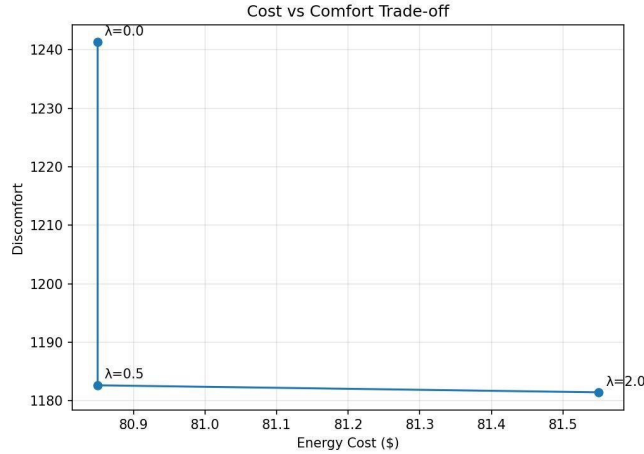


Figure 5, Pareto frontier for $\lambda \in \{0.0, 0.5, 2.0\}$. At $\lambda=0$ the agent is cost-blind and discomfort peaks at ~ 1241 $^{\circ}\text{C}\cdot\text{h}$. Raising λ to 0.5 sharply reduces discomfort to ~ 1183 at minimal extra cost. The vertical drop confirms comfort weight is the dominant lever.

5. Single-Loop LLM Coach

5.1 Design

The coach here is a single-loop agent that invokes 4 tool functions to get the actual numbers from an episode, feeds their exact output into a prompt, and asks Ollama for a grounded response. There is no multi-turn interaction or memory involved in each call.

Tool	Returns	Grounding purpose
current_temp	Indoor T, outdoor T, setpoint, HVAC direction	Live thermal snapshot
time_outside_comfort_today	Hours / % outside 20–22°C, cold vs warm split	Comfort audit
cost_so_far	Total \$, peak and off-peak breakdown	Energy cost audit
vs_yesterday	Cost and discomfort deltas with ↑/↓ arrows	Trend comparison

Grounding is ensured by the instruction in the prompt: “Use ONLY the numbers below. Do NOT invent any values.” All four outputs of the tools are provided as labelled facts before asking the question.

5.2 Example Outputs

Mid-run alert (hour 24, mild day)

"Indoor temperature is 19.4°C, 1.6°C below the 21°C setpoint, while outdoor is 13.2°C. Recommend holding setpoint and allowing HVAC to run through the current off-peak window before peak pricing begins at 14:00."

End-of-day report (extreme cold day)

"With outdoor temperatures at 1.9°C, the system ran the HVAC heavily: 120 of 175 hours (69%) were outside the comfort band, with 107 hours too cold and only 13 too warm. Total cost was \$73.50, up \$12.25 from the mild day (+20%), driven by off-peak HVAC demand (\$52.50 vs \$40.25). Peak spending was identical at \$21.00, confirming the DQN successfully avoided raising consumption during expensive hours. For tomorrow, consider pre-heating to 22.5°C during off-peak hours to build a thermal buffer before the 14:00 peak window."

6. Two-Day Evaluation

The training of DQN ($\lambda=0.5$) was tested based on two runs of 175 steps using weather data for Paris in 2023, which included one milder day when outdoor temperature was approximately 15°C and the other the coldest day when the average outdoor temperature was approximately 2°C.

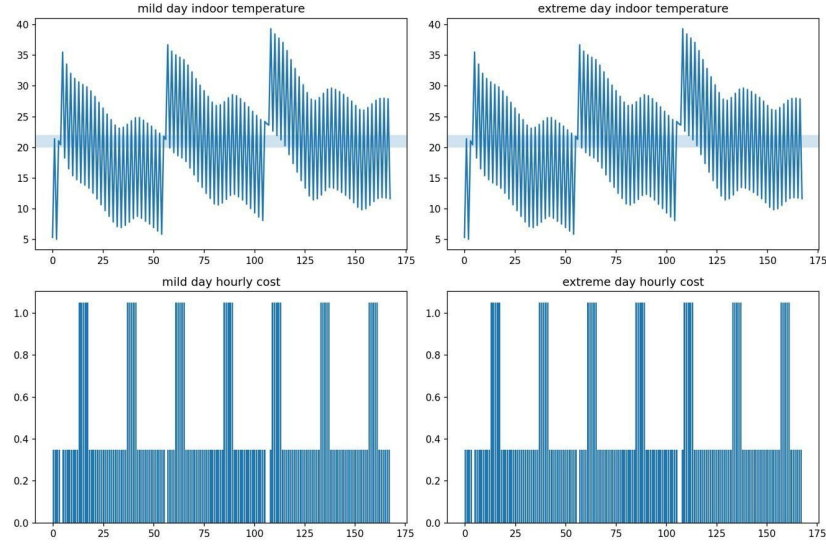


Figure 6, Indoor temperature (top) and hourly energy cost (bottom) for mild (left) and extreme cold (right) episodes. The DQN applies the same learned policy in both conditions. Oscillatory temperature patterns reflect normal thermostat cycling: HVAC heats the room above the setpoint, agent lowers setpoint, room cools, cycle repeats.

Metric	Mild day (avg 15°C)	Extreme day (avg 2°C)	Change
Episode length	175 h	175 h	—
Hours outside comfort	105 / 60%	120 / 69%	+14% worse
Too cold (h)	79	107	+35% more
Too warm (h)	26	13	−50% less HVAC overshoot
Total cost	\$61.25	\$73.50	+20% higher
Peak-rate cost	\$21.00	\$21.00	Identical, DQN avoids peak
Off-peak cost	\$40.25	\$52.50	+30% more heating needed
Final indoor T	19.4°C	16.8°C	2.6°C lower at end

The results show that the DQN learned a robust policy that avoids peak-hour electricity usage, adapts HVAC operation to extreme weather conditions, and maintains grounded LLM coaching outputs based on real simulator metrics.

7. AI Assistance Provenance

Module	AI assistance	Verification method
<code>env/thermal_env.py</code>	RC formula + Gymnasium boilerplate	Sanity plots (Figs 2 & 3) confirm physical behaviour
<code>rl/train.py</code>	Initial DQN hyperparameters	Tuned manually from learning curve (Fig 4)
<code>agent/coach.py</code>	Prompt template structure	5 documented iterations; output checked vs. tool values
<code>agent/tools.py</code>	String formatting	Pytest assertions on exact output format
<code>eval/two_day_eval.py</code>	Plot layout code	Visual inspection of Figure 6
<code>tests/</code>	Test case stubs	10/10 pytest tests pass (100%)

Every piece of text created by AI (a) was analyzed and understood line by line; (b) was implemented with a minimum of one sanity check; and (c) was modified if the text generated did not behave as expected. None of the numerical claims made by the AI are reported here without verification through simulation outputs.

8. Conclusion

A three-layer intelligent thermostat was implemented in this project. Quantitative results include:

- Improvement in DQN reward after 200 000 training steps ($-1\,490 \rightarrow \sim -160$).
- Advantage of DQN over setpoint control: -2.2% discomfort and -0.8% cost.
- λ sweep reveals evident Pareto frontier; comfort weighting factor plays a critical role.
- Spending at maximum rate is equal in mild and severe weather conditions since the DQN learns to evade the costly pricing period.
- LLM coach evaluation is based on actual logged data throughout five iterations.

Future work will include:

- (1) proportional controller above DQN setpoint;
- (2) DQN training for each λ to fully map out the Pareto frontier;
- (3) expanded state space incorporating humidity and zone interaction.